Ankit Kumar

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Summary

Current Robotics M.S. student developing learning-based manipulation, perception, and autonomous systems. Experienced with real robots, imitation learning, diffusion policies, stereo vision, SLAM, ROS 2, PyTorch, and embedded deployment.

Education

University of Minnesota — Twin Cities, MN
M.S. in Robotics, GPA: 3.78/4.0 — Sep 2024 - Dec 2026
Coursework: Intelligent Robotic Systems, Robot Vision, Machine Learning, Deep Learning, Computer Vision

Jadavpur University — Kolkata, India
B.E. in Mechanical Engineering, GPA: 8.55/10 — Aug 2018 - May 2022

Technical Skills

Programming & Tools: Python, C++, MATLAB, Bash, Linux, Git
Robot Learning: Imitation Learning, Behavior Cloning, ACT, Diffusion Policy, Vision-Language-Action (VLA) Models
Computer Vision & ML: PyTorch, OpenCV, U-Net, YOLO, Stereo Vision, RGB-D Perception, Semantic Segmentation, Object Tracking
Robotics & Autonomy: ROS/ROS 2, Nav2, RTAB-Map, AMCL, LiDAR, Sensor Fusion, Dijkstra's Algorithm
Simulation & Hardware: NVIDIA Isaac Sim, Gazebo, ManiSkill, SO-101, NVIDIA Jetson Orin Nano
Aerial Robotics: PX4, ArduPilot, MAVROS, RTK-GNSS, Camera Calibration

Experience

Flow Field Imaging Lab, University of Minnesota — Twin Cities, MN
Graduate Research Assistant — May 2025 - Present

Autonomous UAV Perception & 3D Ember Tracking

- Developed an end-to-end **perception-driven UAV autonomy pipeline**, integrating synchronized stereo cameras, NVIDIA Jetson Orin Nano, and onboard flight control for real-time wildfire ember tracking.
- Designed and trained a **multi-head U-Net with self-attention** and dedicated mask, centerline, and endpoint prediction heads, achieving **90.3% IoU** for ember segmentation under challenging visual conditions.
- Implemented stereo calibration, correspondence matching, triangulation, and temporal association for ember trajectory reconstruction, achieving **<10 cm 3D error** in controlled tests.
- Optimized embedded inference to **approximately 10 FPS** on Jetson Orin Nano and integrated perception outputs with the UAV control loop for autonomous response to observed ember motion.

Multi-UAV Autonomous Field Deployment

- Built and field-tested a **four-UAV leader-follower system** using ROS 2, MAVROS, PX4, and RTK-GNSS, coordinating waypoint missions and GPS-based formation offsets during prescribed-burn experiments.
- Integrated synchronized imagery, GNSS, and telemetry across four UAVs for multi-point smoke-dispersion data collection.

Projects

Vision-Based Robotic Manipulation | SO-101, ACT, Diffusion Policy, PyTorch

- Built a vision-conditioned manipulation pipeline with fixed and wrist cameras, collecting **100 teleoperated demonstrations** with synchronized images, robot joint states, and actions.
- Trained and evaluated **ACT and Diffusion Policy** using **85 training and 15 held-out validation episodes**, with Diffusion Policy achieving **31% lower arm-joint prediction error** and **34% lower gripper prediction error** than ACT.
- Deployed both policies on the physical SO-101 robot for autonomous, vision-conditioned cube-to-cup manipulation.

Autonomous Mobile Robot Navigation & SLAM | ROS 2, RTAB-Map, Nav2, LiDAR

- Developed an autonomous mobile-robot pipeline in **NVIDIA Isaac Sim**, integrating RGB-D SLAM, LiDAR-based localization, occupancy-grid mapping, and Nav2 waypoint navigation.
- Implemented Dijkstra path planning and differential-drive control, evaluating localization accuracy with **0.82 m ATE RMSE** and **0.19 m RPE RMSE** using evo.
- Diagnosed TF alignment, synchronization, and map-consistency issues to improve navigation reliability.

NeRF-Based 3D Reconstruction | OpenCV, PyTorch, COLMAP, Instant-NGP

- Built a drone-image reconstruction pipeline using **COLMAP and Instant-NGP**, incorporating autonomous viewpoint selection for 3D scene reconstruction.
- Applied image-quality filtering and target-region cropping to improve camera-pose estimation and reconstruction fidelity.

Autonomous VTOL Flight Simulation | Gazebo, ROS 2, MAVROS, ArduPilot

- Developed a **Gazebo-ArduPilot SITL** environment to validate autonomous VTOL waypoint missions.
- Integrated MAVROS command and telemetry interfaces for repeatable mission testing and flight-state monitoring.